

Nguyen Thuy Trang

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Objective

With a background in Mathematics and Computer Science, I am interested in building AI-powered and data-driven systems that solve practical problems. I enjoy developing solutions that improve workflows, support better decision-making, and create meaningful real-world impact. I am especially motivated by projects that combine technical problem-solving with a strong understanding of user needs, and I look forward to contributing to collaborative teams where I can continue learning, build useful products, and turn ideas into practical solutions.

Work Experience

- Institute of Physics - VAST**, Research Assistant InternMay - August 2025
- Processed and cleaned multi-modal sensor signals; extracted features for classification tasks; Researched AI/ML approaches for analyzing sensor data in physics and biomedical applications.
 - Assisted in academic writing and preparation of research reports.
- Dasens**, DeveloperFeb 2026 – present
- Collaborated remotely with a Canada-based cross-functional team to develop an AI-powered EdTech platform for personalized student assessment and learning analytics.
 - Developed backend services and contributed to AI-driven features, including retrieval and workflow automation.
 - Technologies: Python, Node.js, React.js, LangChain, LightRAG, n8n.

Education

- VNU - University of Science**, BS in Mathematics and Computer Science2022 – now
- CPA: 3.3/4.0
 - **Relevant Coursework:** Algebra, Calculus, Probability, Statistics, Data Structures & Algorithms (DSA), Machine Learning, Computer Vision, Natural Language Processing & Apps.

SKILLS

Programming Languages	Python, Java, JavaScript, MATLAB, R, Arduino
Machine Learning & AI	Deep Learning, NLP, LLMs, RAG, Knowledge Graph, Agentic AI
Data	SQL, relational database, Spark
Tools & Technologies	Git/GitHub, Docker, Notion, Latex, AI Tools, n8n

LANGUAGES

English	CEFR B2
Japanese	JLPT N4 (currently preparing for N3)

PROJECTS

- HomeMind – AI-powered Smart Home Management Application**2026
- Developed a smart home management application that enables users to monitor and control IoT devices such as lighting, air conditioners, smart locks, cameras, and environmental sensors through a centralized interface.
 - Features: Real-time device monitoring and control; room-based management; automation routines and scheduling; sensor-based triggers; notifications for abnormal events; energy consumption tracking.
 - AI Agent: Integrated an LLM-based assistant for natural-language device control, automatic routine creation, smart-home status queries, and context-aware recommendations based on sensor data and user preferences.
 - Intelligent automation: Combined environmental data and user behavior to automatically adjust devices, suggest energy-saving actions, and execute multi-device scenarios such as “Good Night” or “Leaving Home”.

- Technologies: React/Flutter, FastAPI, MQTT, WebSocket, PostgreSQL, Docker, LLM, LangGraph, Langfuse, ZEP.

AYALEARNING – AI-Powered Japanese Learning Platform

2025

- Architected and developed a full-stack AI-powered language learning platform integrating dictionary search, adaptive vocabulary training, interactive practice games (inspired from traditional Japanese games), and real-time conversational AI with animated Live2D characters, enabling users to practice kaiwa with different AI-driven personalities through immersive avatar interaction.
- Implemented a personal AI assistant agent using LangChain and LangGraph to orchestrate multi-step task execution, including answering questions, generating personalized study plans, scheduling reviews, and guiding user workflows across the platform.
- Technologies: ReactJS, TypeScript, TailwindCSS, NodeJS, PostgreSQL, LangGraph, LLM APIs, Live2D.

Gardenia - FlowerVision

2025

- Developed a web-based flower recognition application using Deep Learning, combining ConvNeXt for flower classification and YOLOv8 for automatic flower detection and cropping.
- Implemented real-time image recognition through webcam and image upload, supporting top-3 predictions with confidence scores, large image preview, drag-and-drop upload, and responsive UI design.
- Built a Flask backend with API endpoints for image classification and real-time detection. The inference pipeline follows: input image, YOLO detection, cropped flower region, ConvNeXt classification, and Wikipedia information retrieval to provide detailed flower information.
- Field: Computer Vision, Deep Learning, Image Classification, Object Detection, Web Application Development.

RESEARCH EXPERIENCES

Application of machine learning in knee-injury detection based on IMU sensor data

2025

The 9th International Conference on Applied & Engineering Physics

Nguyen Thuy Trang, Pham Thi Duyen, Nguyen Trong Hieu, Nguyen Quang Linh, Nguyen Anh Duong, Vu Duong

Building a dataset and information retrieval system from admission counseling forums

2025

Third Prize, Annual Student Research Conference, MIM, HUS – Apr 2025

Pham Tuan Anh, **Nguyen Thuy Trang**, Le Thanh Truong, Nguyen Cong Manh

CERTIFICATES & AWARDS

The 9th Global Competition for Future Design, Southeast Asia Regional Finals

July 2026

Best Technologically Impressive Award

Global Program on Innovation and Entrepreneurship

Sep - Nov 2025

Tsinghua University, China

Guangxi University International Students AI Winter Camp 2025

Dec 2025

Guangxi University, China

VIASM Summer School “Research Experience for Undergraduates”

2025

Vietnam Institute for Advanced Studies in Mathematics, Vietnam

Summer School of Agentic AI

2025

Naver & Hanoi University of Science and Technology (HUST)

Samsung Innovation Campus - AI Track

2024

EXTRACURRICULAR ACTIVITIES

HUS Applied Mathematics and Informatics Club, Head of Events

2023 – 2025

ULIS Japanese Club, member

2023 – 2025

Math Open Day, volunteer

2024